

Brain tumor segmentation

Introduction

Brain tumors represent one of the most serious neurological disorders, requiring precise diagnosis and accurate localization to support effective treatment planning and improve patient survival. Magnetic Resonance Imaging (MRI) has become the standard imaging modality for brain tumor assessment due to its superior soft-tissue contrast and ability to reveal detailed anatomical structures. Despite its clinical importance, manual segmentation of brain tumors remains a labor-intensive and subjective task that is highly dependent on radiologists' expertise, making it susceptible to variability and increased diagnostic workload.

Recent advances in Artificial Intelligence, particularly Deep Learning, have significantly transformed the field of medical image analysis by enabling automated and highly accurate segmentation of complex anatomical structures. Semantic segmentation models can identify tumor boundaries at the pixel level, providing detailed quantitative information that supports clinical decision-making while reducing human effort.

In this project, we developed a comprehensive AI-powered brain tumor segmentation framework based on the U-Net++ architecture with an EfficientNet encoder. The proposed framework was designed as a complete end-to-end pipeline, starting from MRI image preprocessing and advanced data augmentation, followed by model training using transfer learning and multiple optimization strategies, and ending with extensive evaluation and automated result generation.

To maximize segmentation performance and improve model generalization, the system incorporates several state-of-the-art techniques, including hybrid loss functions, mixed precision training, gradient accumulation, CutMix augmentation, Exponential Moving Average (EMA), Test-Time Augmentation (TTA), and Grad-CAM-based explainability. Furthermore, the framework performs comprehensive post-hoc analysis by generating quantitative evaluation metrics, visual comparisons between predictions and ground truth masks, error analysis, and automatically exported reports for reproducibility and further clinical assessment.

The developed system demonstrates how modern deep learning techniques can provide an efficient, accurate, and interpretable solution for automated brain tumor segmentation from MRI images. By reducing the dependence on manual delineation while maintaining high segmentation quality, the proposed framework has the potential to support radiologists in clinical workflows and contribute to faster, more consistent, and data-driven medical decision-making.

Phase 1: Tumor Detection (Production Model)

The first phase focuses on developing a **production-ready classification model** that predicts whether an MRI scan contains a brain tumor or not.

The model is trained to distinguish between **healthy brain tissues** and **tumor-affected tissues**, regardless of the tumor's type, size, shape, or appearance. This binary classification serves as the initial screening stage, enabling the system to quickly identify MRI scans that require further analysis in the subsequent segmentation phase.

```
Epoch 17/30  LR_head=2.24e-05

Train  loss=0.2829  recall=0.7476
Val    loss=0.1735  recall=0.9889  auc=0.9974  composite=0.9914
No improvement (6/8)  best=0.9943

Epoch 18/30  LR_head=1.99e-05

Train  loss=0.2990  recall=0.7317
Val    loss=0.1739  recall=0.9873  auc=0.9973  composite=0.9903
No improvement (7/8)  best=0.9943

Epoch 19/30  LR_head=1.73e-05

Train  loss=0.2952  recall=0.7340
Val    loss=0.1710  recall=0.9905  auc=0.9968  composite=0.9924
No improvement (8/8)  best=0.9943

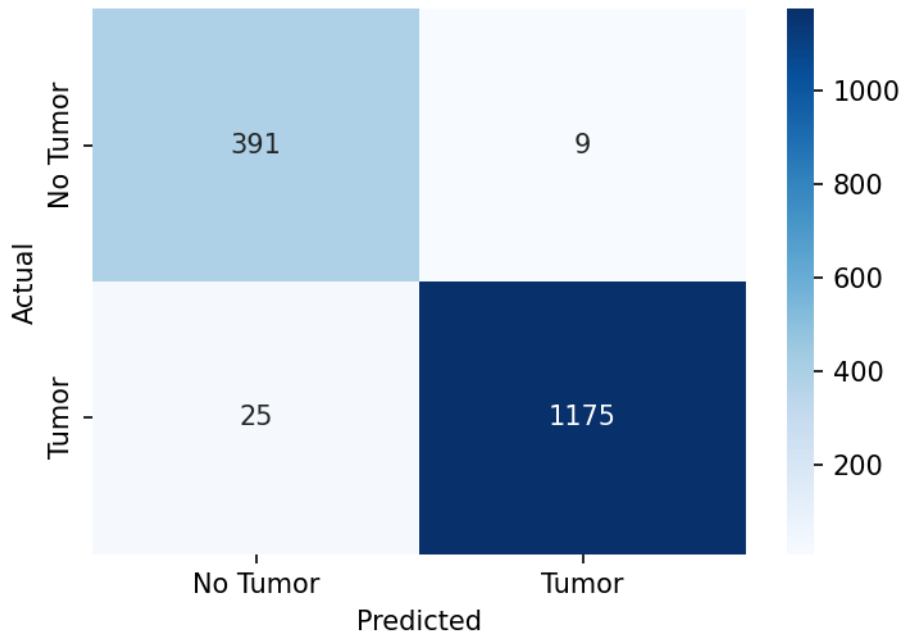
Early stopping – best at epoch 11
```

Here Model Performance

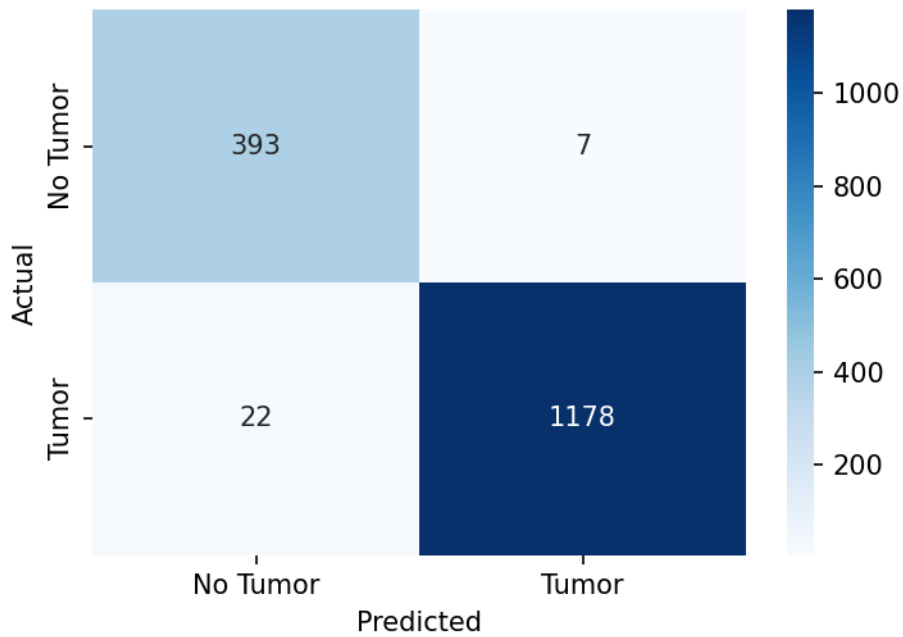
After training, the model achieved high classification accuracy, demonstrating its ability to reliably distinguish between **healthy brain MRI scans** and **tumor-containing scans**.

The achieved accuracy reflects the model's effectiveness in identifying the presence or absence of a brain tumor, making it a reliable first-stage screening model before proceeding to detailed tumor segmentation.

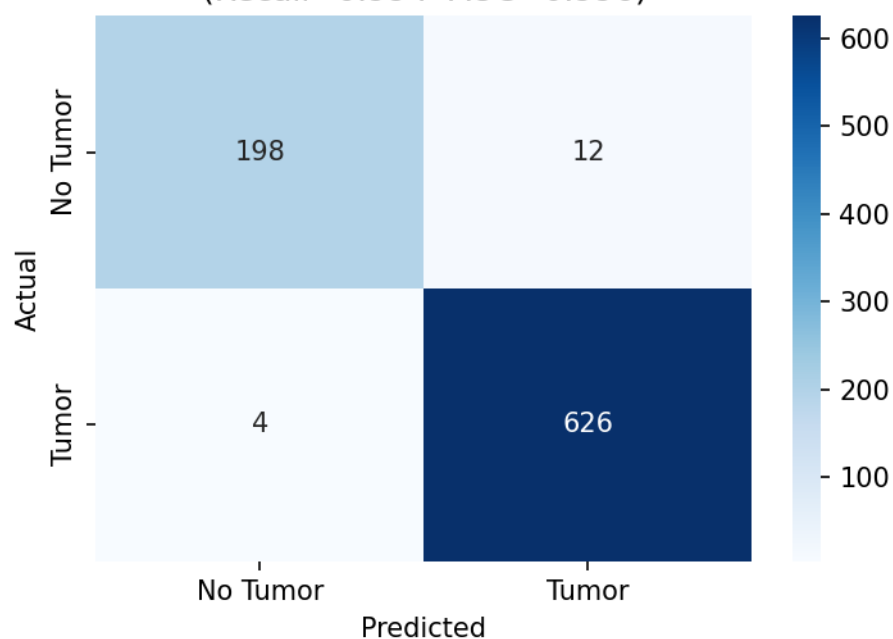
Test Set - Confusion Matrix
(Recall=0.979 AUC=0.997)



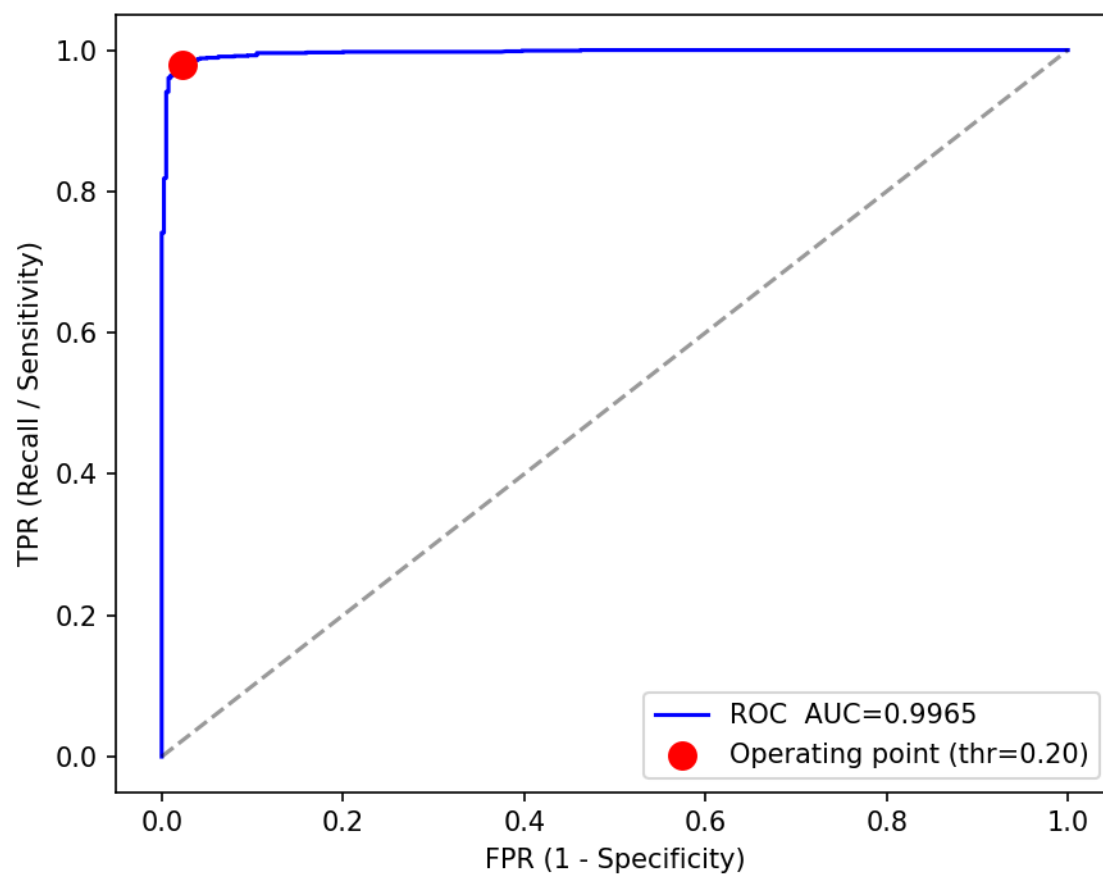
Test Set TTA - Confusion Matrix
(Recall=0.982 AUC=0.998)



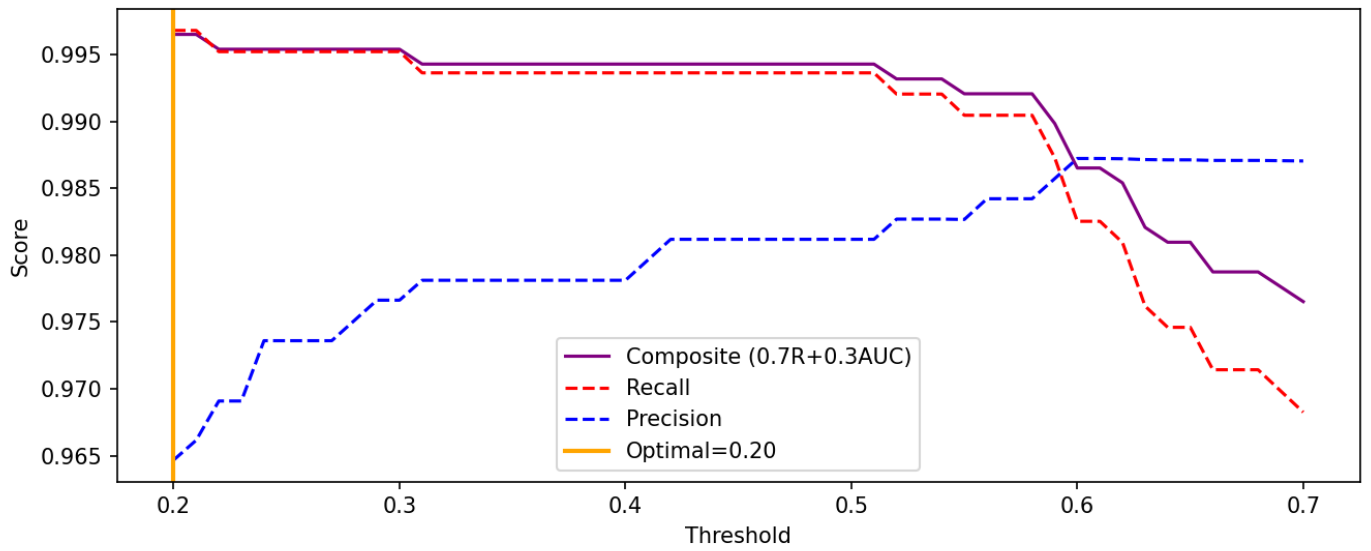
Validation Set - Confusion Matrix
(Recall=0.994 AUC=0.996)



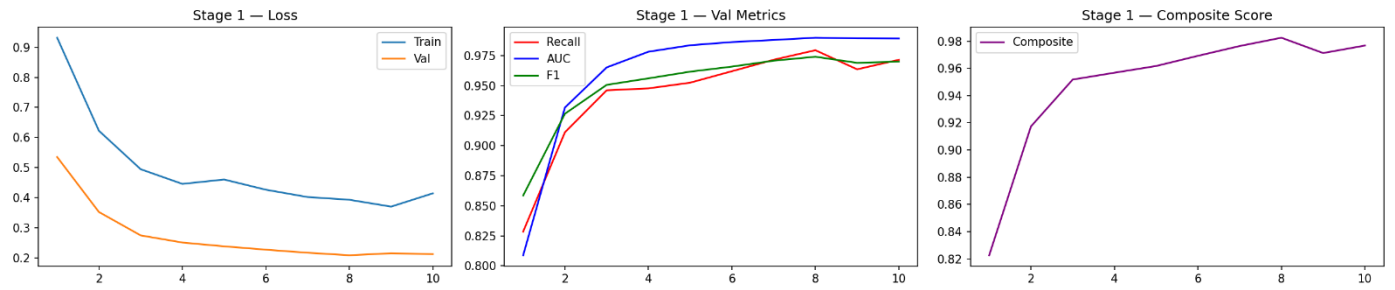
ROC Curve - Test Set



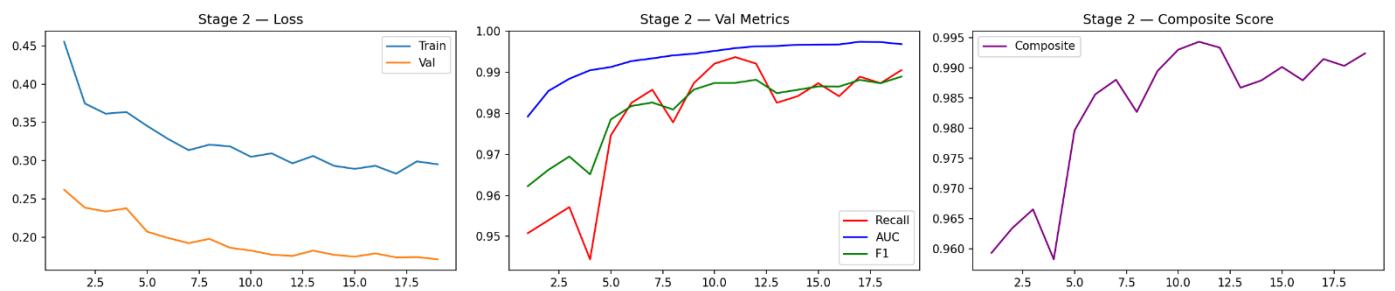
Threshold Calibration on Validation Set



Training History — Stage 1



Training History — Stage 2



Phase 2: Multi-Class Tumor Classification

CLASSIFICATION CLASSES

```
Train classes: ['pituitary', 'no_tumor', 'meningioma', 'glioma']  
Val classes   : ['pituitary', 'no_tumor', 'meningioma', 'glioma']
```

The second phase focuses on training the model to classify different brain tumor categories using the **EfficientNet-B3** architecture.

During training, an issue was observed where the **validation loss intermittently became NaN**, despite the training process continuing normally. To investigate the problem, we verified the number of classes before and after training and carefully reviewed the dataset and preprocessing pipeline. Since the training and validation accuracy remained stable, we concluded that the issue was not caused by corrupted data or preprocessing errors.

Further analysis revealed that the instability originated from the combination of **Focal Loss** and **Automatic Mixed Precision (AMP)**. AMP accelerates training by performing part of the computations in **float16** precision, which is more susceptible to numerical instability. Within the Focal Loss computation, the exponential term $\exp(-\text{CE Loss})$ can become numerically unstable when the cross-entropy loss is extremely large or extremely small, particularly during the early training stages when the backbone is still frozen.

As a result, only specific validation batches experienced numerical overflow or underflow, causing the validation loss to become **NaN** in some epochs while remaining normal in others. Since this behavior occurred only for certain batches—and not consistently across the entire dataset—it confirmed that the problem was related to numerical precision rather than data quality or model architecture.

```
=====
0.4300/nan  Ac
0.3915/0.7682
0.3759/nan  Ac
0.3619/nan  Ac
0.3812/0.1922
0.3613/0.1668
0.3484/0.1480
0.3575/nan  Ac
0.3289/1.0086
0.3553/0.5904
0.3447/nan  Ac
0.3268/0.1298
0.3295/0.1814
0.3519/nan  Ac
0.3102/nan  Ac
0.3174/1.5234
0.3280/0.1213
0.3233/0.2112
0.3166/0.1407
0.3212/0.1787
```

During the development of the multi-class classification model, numerical instability was observed during validation, resulting in intermittent **NaN validation loss** despite stable training accuracy.

A detailed investigation confirmed that the dataset, preprocessing pipeline, and class distribution were all correct. The issue was traced to numerical instability caused by the interaction between mixed-precision training and the selected loss function rather than a problem with the data itself.

Solution

To overcome this limitation, the classification pipeline was redesigned by integrating the classification task into a more robust **multi-task learning framework**.

Instead of treating classification as an isolated task, the final architecture learns **tumor classification and tumor segmentation simultaneously**, enabling the model to extract richer spatial and semantic features from MRI images. This redesign improves training stability, enhances feature representation, and produces more reliable classification performance while supporting accurate tumor localization.

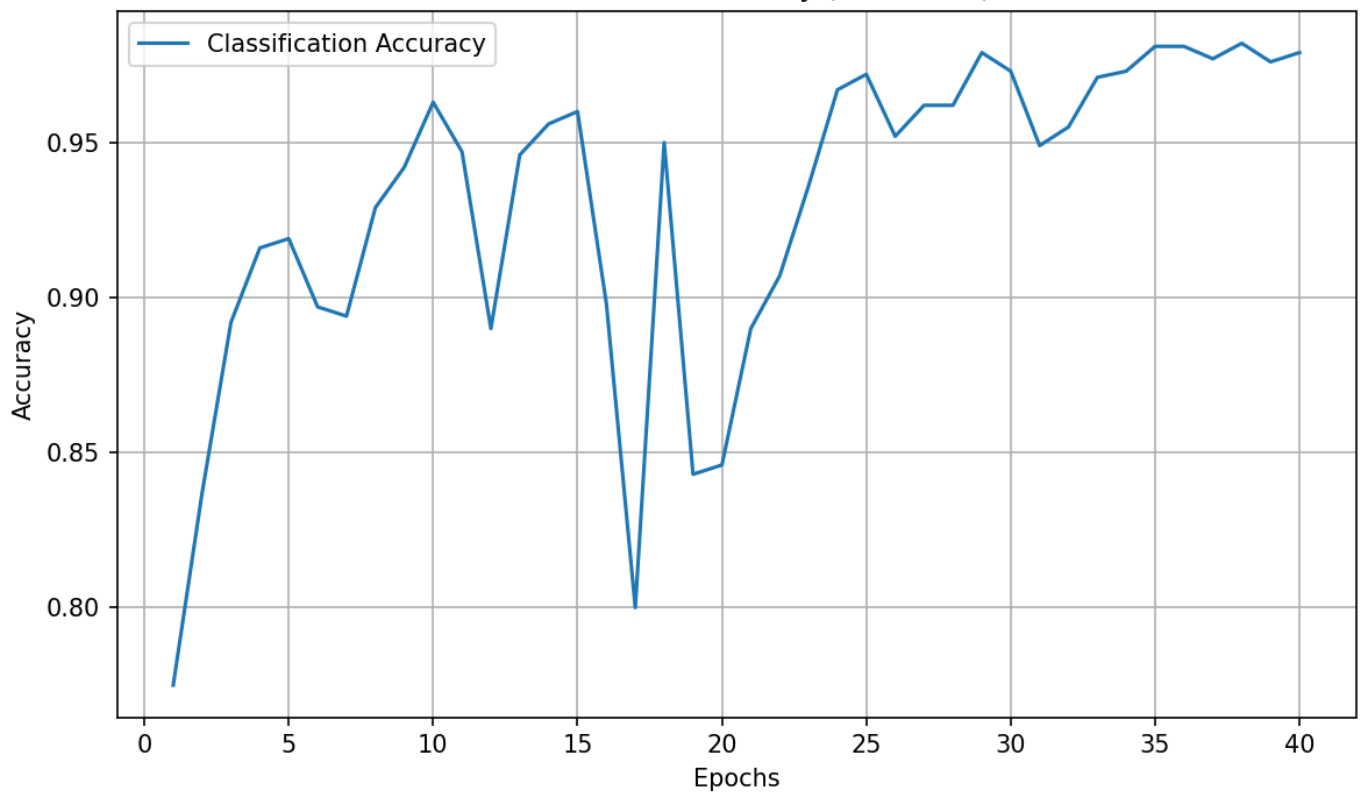
Classification Results

The final classification model successfully learned meaningful MRI features that differentiate between tumor classes with high confidence.

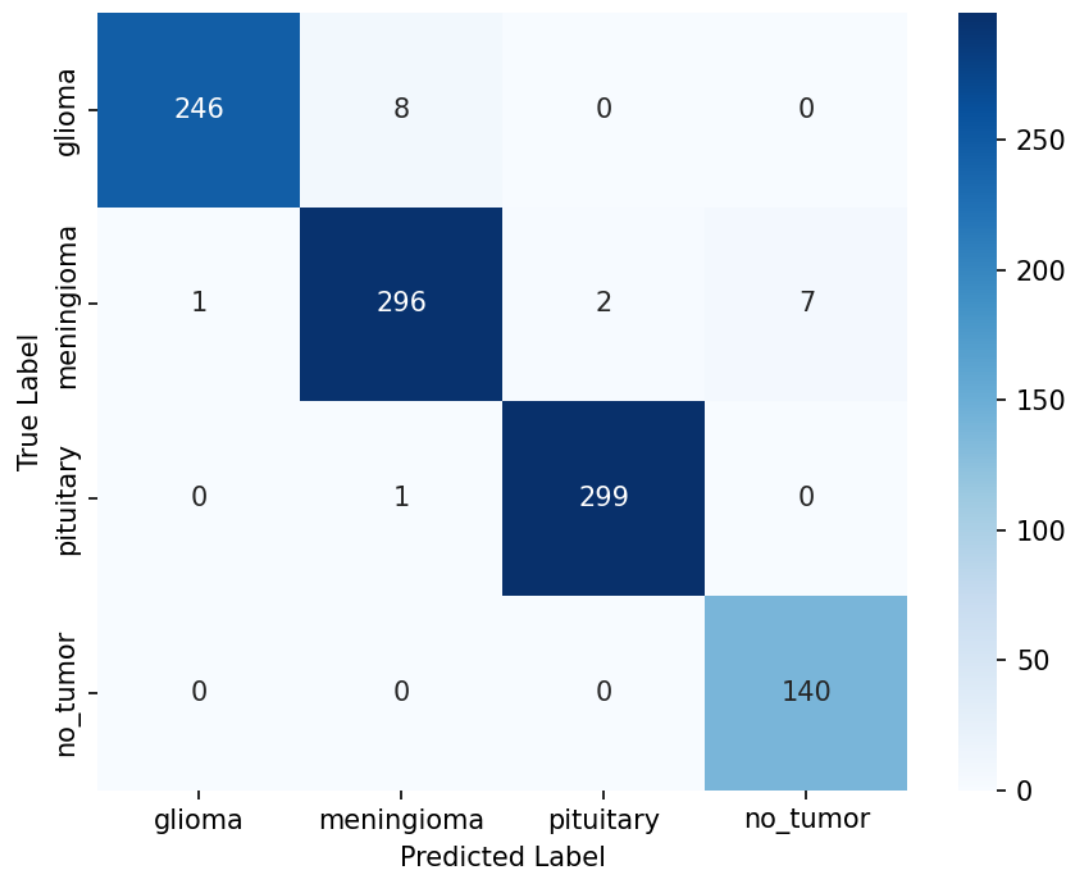
The learned representations are subsequently utilized by the segmentation network, providing a strong foundation for accurate tumor localization and boundary delineation.

```
Epoch [40/40]
Segmentation Train: 100%|██████████| 394/394 [00:40<00:00, 9.75it/s, seg_loss=0.127]
Classification Train: 100%|██████████| 500/500 [00:31<00:00, 16.05it/s, cls_loss=0.000121]
Seg Loss      : 0.1028
Cls Loss      : 0.0029
Dice Score    : 0.8407
IoU Score     : 0.7330
Accuracy      : 0.9790
Combined      : 0.8509
```

Classification Accuracy (Validation)



Classification Confusion Matrix



Model 3 segmentation model

Why Segmentation?

Detecting the presence of a brain tumor is only the first step. For clinical diagnosis and treatment planning, it is equally important to determine the **exact location, shape, and boundaries** of the tumor.

Unlike image classification, which provides a single label for the entire MRI scan, segmentation performs **pixel-level analysis**, allowing the model to identify every pixel that belongs to the tumor region. This produces a precise tumor mask that can assist clinicians in diagnosis, surgical planning, and treatment monitoring.

What is Image Segmentation?

Image segmentation is the process of dividing an image into meaningful regions by assigning a class label to every individual pixel.

In our project, each MRI image is transformed into a corresponding segmentation mask, where every pixel is classified as either **tumor tissue** or **healthy brain tissue**. The final output is a pixel-level map that accurately highlights the tumor boundaries while preserving the surrounding anatomical structures.

Why Pixel-Level Prediction?

Pixel-level segmentation provides significantly more detailed information than image classification alone.

Instead of simply answering "**Is there a tumor?**", the model determines:

- Where the tumor is located.
- Which pixels belong to the tumor.
- The exact tumor shape.
- The tumor boundaries.
- The tumor size and occupied area.

This level of precision is essential for medical image analysis and clinical decision support.

From Classification to Segmentation

The segmentation stage builds upon the previous classification pipeline.

The overall workflow follows a hierarchical decision process:

- **Stage 1:** Detect whether a tumor is present.
- **Stage 2:** Identify the tumor category.
- **Stage 3:** Precisely localize the tumor by generating a pixel-wise segmentation mask.

This progressive pipeline enables the system to move from simple image-level understanding to detailed anatomical localization.

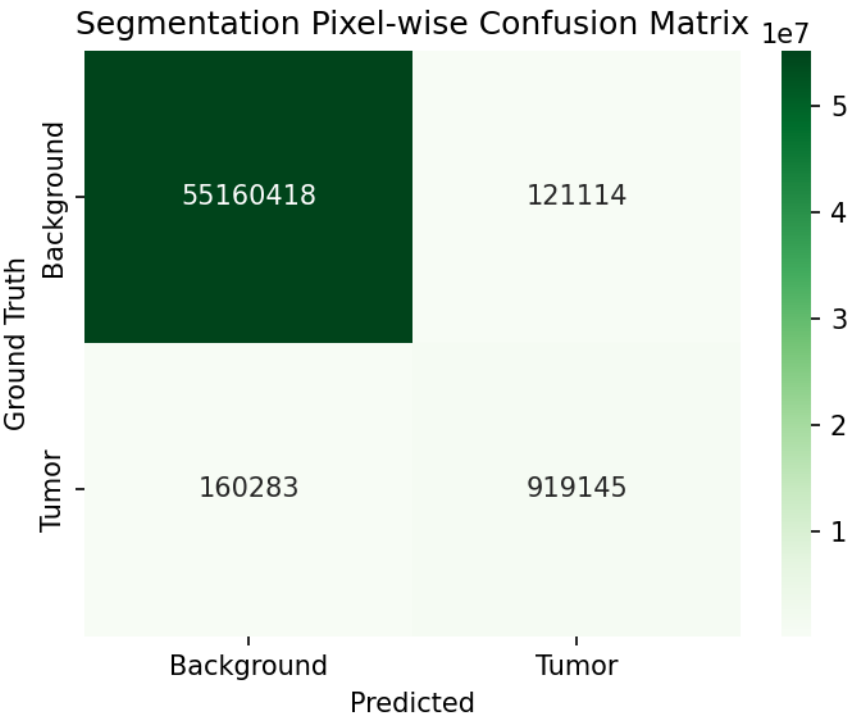
Segmentation Strategy

To accurately segment brain tumors from MRI images, we adopted a deep learning-based semantic segmentation approach.

The model learns both **global contextual information** and **fine-grained spatial details**, enabling it to distinguish tumor tissue from healthy brain structures even when tumor shapes, sizes, and appearances vary considerably.

Rather than relying on handcrafted image processing techniques, the network automatically learns hierarchical visual features directly from the MRI data, resulting in robust and accurate tumor localization.

```
Epoch [40/40]
Segmentation Train: 100%|██████████| 394/394 [00:40<00:00, 9.75it/s, seg_loss=0.127]
Classification Train: 100%|██████████| 500/500 [00:31<00:00, 16.05it/s, cls_loss=0.000121]
Seg Loss      : 0.1028
Cls Loss      : 0.0029
Dice Score    : 0.8407
IoU Score     : 0.7330
Accuracy      : 0.9790
Combined      : 0.8509
```



Final AI Pipeline

End-to-End Brain Tumor Analysis System

Our project was designed as a complete **AI-powered diagnostic pipeline**, where each stage builds upon the output of the previous one to provide a comprehensive analysis of brain MRI scans.

The workflow begins by receiving an MRI image as input. The first stage performs **binary classification** to determine whether the scan contains a brain tumor. If no tumor is detected, the system immediately reports a negative diagnosis.

When a tumor is detected, the MRI image is forwarded to the **multi-class classification model**, which identifies the specific tumor type. This provides more detailed diagnostic information and prepares the image for further analysis.

The next stage performs **semantic segmentation**, where the tumor is localized at the pixel level. Instead of simply identifying the presence of a tumor, the segmentation model precisely outlines its boundaries, allowing accurate localization and shape reconstruction.

Using the generated segmentation mask, the system calculates quantitative tumor characteristics, including the **tumor size, occupied area, and estimated dimensions**, providing clinically meaningful measurements that support medical assessment.

To integrate all AI components into a unified framework, the trained models are deployed using their saved **H5 weight files**, while **JSON configuration files** define each model's architecture and enable seamless loading within the inference pipeline.

Finally, the outputs from all models—including tumor detection, tumor classification, segmentation results, and quantitative measurements—are forwarded to a **Large Language Model (LLM)** through an API. The LLM analyzes these structured results and automatically generates a comprehensive medical report that summarizes the diagnosis, tumor type, localization, and quantitative findings in a clear and readable format.

The result is a complete AI-assisted diagnostic system that transforms a single MRI scan into an interpretable clinical report, supporting radiologists with faster, more consistent, and data-driven decision-making.